# Department of Electronics and Electrical Engineering
# Indian Institute of Technology Guwahati

## EE101 - Basic Electronics

**Tutorial 3: Solutions** **Date: Aug 24, 2023**

1. Consider the electrical circuit shown in Figure 1. Evaluate the voltage $v_x$ across the resistor $R_3$ using nodal analysis. Use the following numerical values of the circuit elements

$$V_1 = 100V,\ V_2 = 150V, I_1 = 5A,\ I_2 = 10A$$
$$R_1 = 20\Omega,\ R_2 = 10\Omega,\ R_3 = 25\Omega,\ R_4 = 12.5\Omega$$

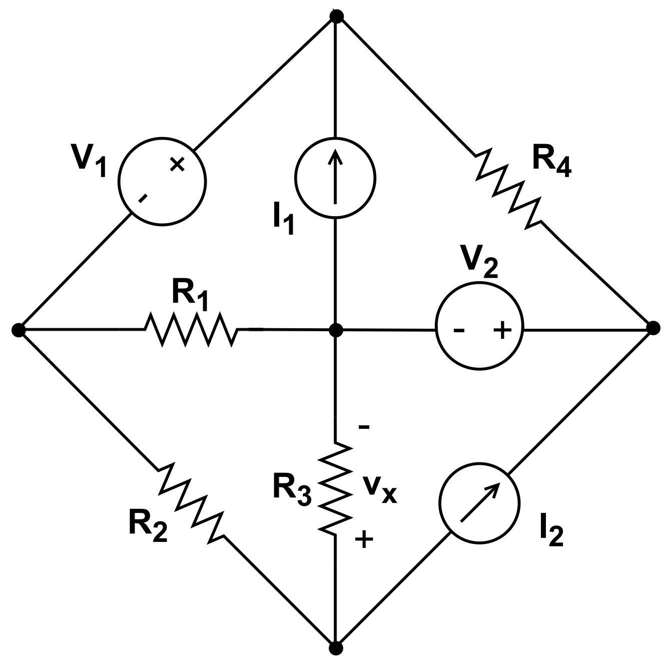


Figure 1 Electrical circuit with independent voltage and current source. Evaluate $v_x$.

**Solution**

Refer to Figure 2. There is an independent voltage source $V_1$ connected between nodes A and B. Thus, we can write $v_b - v_a = V_1$ or $v_b = v_a + V_1$

A super-node is formed by combining nodes A and B. The current equation at this super-node gives us the following

$$G_2(v_x - v_a) - G_1 v_a + I_1 - G_4(v_b - v_c) = 0 \tag{1}$$

$$\text{Now, } v_b = v_a + V_1 \text{ and } v_c = V_2$$

This substitution gives us

$$G_2(v_x - v_a) - G_1 v_a + I_1 - G_4(v_a + V_1 - V_2) = 0$$

Rearranging terms, we obtain

$$(G_1 + G_2 + G_4)v_a - G_2 v_x = I_1 - G_4(V_1 - V_2)$$

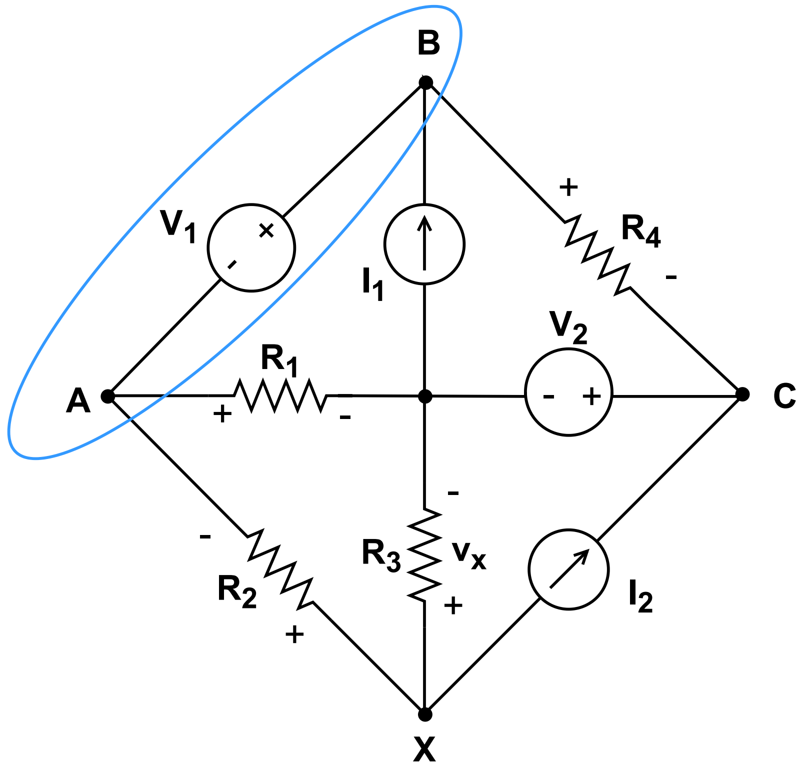


Figure 2 The nodes are marked as A,B,C and X. Node R is taken as reference node. Nodes A and B are combined to form a super-node.

Putting the numerical values of circuit elements, we obtain

$$0.23v_a - 0.1v_x = 9 \tag{2}$$

KCL at node X gives us

$$-G_2(v_x - v_a) - G_3v_x - I_2 = 0$$

Rearranging terms, we get

$$-G_2v_a + (G_2 + G_3)v_x = -I_2$$

Putting the numerical values of circuit elements, we obtain

$$-0.1v_a + 0.14v_x = -10 \tag{3}$$

Eliminating $v_a$ from equations (1) and (2), we get the value of $v_x$ as

$$v_x = -63.06V$$

2. Consider the electrical circuit shown in Figure 3. Evaluate the mesh currents $i_1$, $i_2$, $i_3$ and the voltages $V_{I1}$, $V_{I2}$ across the respective current sources $I_1$ and $I_2$. Use the following numerical values of the circuit elements.

$$V_1 = 4V,\ V_2 = 5V,\ V_3 = -11V$$
$$I_1 = k_1 \times i_x = 2i_x,\ I_2 = k_2 \times v_x = \tfrac{v_x}{12}$$
$$R_1 = 2\Omega,\ R_2 = 3\Omega,\ R_3 = 1\Omega,\ R_4 = 1\Omega,\ R_5 = 4\Omega$$

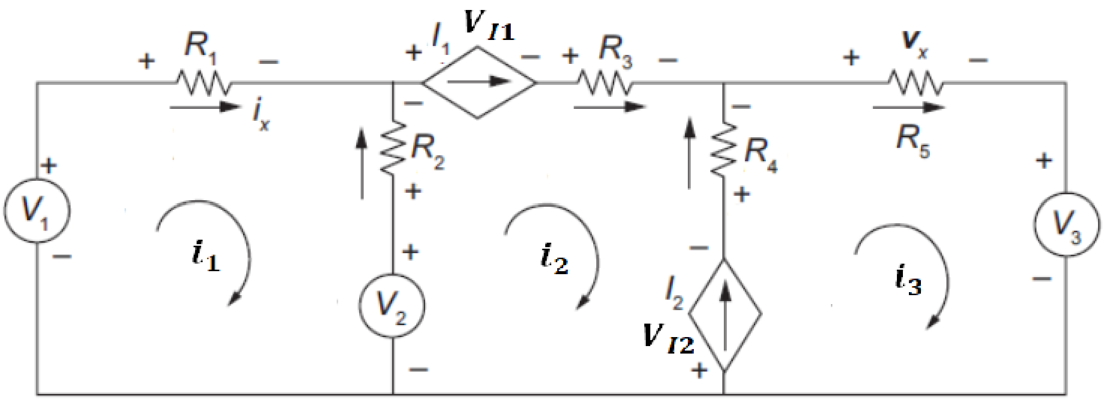


Figure 3 Electrical circuit with dependent and independent sources.

**Solution**

From mesh 1, we observe that $i_1 = i_x$
Also, in mesh 2, we observe that

$$i_2 = I_1 = k_1 i_x = 2i_1 \tag{4}$$

Application of KVL in mesh 1 gives us

$$-V_1 + i_1 R_1 - R_2(i_2 - i_1) + V_2 = 0$$
$$i_1 R_1 - R_2(2i_1 - i_1) = V_1 - V_2$$

Using the numerical values, we obtain the following

$$i_1 = 1A$$
$$\text{Thus, } i_2 = 2A$$

From mesh 3, We observe that

$$i_3 - i_2 = I_2 = k_2 v_x = k_2 i_3 R_5$$
$$=> i_3 = i_2 + k_2 R_5 i_3$$

Using numerical values, we obtain

$$i_3(1 - \frac{1}{12} * 4) = 2$$
$$=> i_3 = 3A$$

KVL in mesh 3 gives

$$V_{I2} + R_4(i_3 - i_2) + R_5 i_3 + V_3 = 0$$

Using the numerical values, we get

$$V_{I2} = -2V$$

KVL in mesh 2 gives

$$-V_2 + R_2(i_2 - i_1) + V_{I1} + R_3 i_2 - R_4(i_3 - i_2) - V_{I2} = 0$$

Using the numerical values, we get

$$V_{I1} = -1V$$

3. The following questions relate to base 2, base 3, base 8, base 10 and base 16 representation of numbers. Note that the alphabet for base 16 is 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F.

(a) Compute the base conversions as specified in the table:

| Base | Number | New Base | Number |
|---|---|---|---|
| 10 | 65/7 | 2 | |
| 10 | 72/7 | 3 | |
| 8 | 4567 | 16 | |
| 16 | 10AF | 8 | |

(b) Find the 4 most significant bits in the base 2 representation of $(\sqrt{2})_{10}$.

**Solution**

(a) $(65/7)_{10} = (9 + 2/7)_{10}$. By repeatedly dividing by 2 and finding remainders we get $(9)_{10} = (1001)_2$. By repeatedly multiplying by 2 and finding the integer parts we get $(2/7)_{10} = (.01\underline{001})_2$. Thus $(65/7)_{10} = (1001.01\underline{001})_2$.

$(72/7)_{10} = (10 + 2/7)_{10}$. By repeatedly dividing by 3 and finding remainders we get $(10)_{10} = (101)_3$. By repeatedly multiplying by 3 and finding the integer parts we get $(2/7)_{10} = (.\underline{021201})_3$. Thus $(72/7)_{10} = (101.\underline{021201})_3$.

$(4567)_8 = 4.8^3 + 5.8^2 + 6.8^1 + 7.8^0 = (1.2^2 + 0.2^1 + 0.2^0).(2^3)^3 + (1.2^2 + 0.2^1 + 1.2^0).(2^3)^2 + (1.2^2 + 1.2^1 + 0.2^0).(2^3)^1 + (1.2^2 + 1.2^1 + 1.2^0).(2^3)^0 = (1.2^2 + 0.2^1 + 0.2^0).(2^9) + (1.2^2 + 0.2^1 + 1.2^0).(2^6) + (1.2^2 + 1.2^1 + 0.2^0).(2^3) + (1.2^2 + 1.2^1 + 1.2^0).(2^0) = (1.2^{11} + 0.2^{10} + 0.2^9) + (1.2^8 + 0.2^7 + 1.2^6) + (1.2^5 + 1.2^4 + 0.2^3) + (1.2^2 + 1.2^1 + 1.2^0) = (100\ 101\ 110\ 111)_2 = (100101110111)_2$.
In short form $(4567)_8 = (100\ 101\ 110\ 111)_2 = (100101110111)_2$. Similarly, $(100101110111)_2 = (1001\ 0111\ 0111)_2 = (9\ 7\ 7)_{16} = (977)_{16}$.

$(10AF)_{16} = (0001\ 0000\ 1010\ 1111)_2 = (1000010101111)_2 = (001\ 000\ 010\ 101\ 111)_2 = (1\ 0\ 2\ 5\ 7)_8 = (10257)_8$.

(b) As $1 < 2 < 4$, we have $1 < \sqrt{2} < 2$. Thus the integer part of $\sqrt{2}$ is 1. This is represented in base 2 as 1. The fractional part is $\sqrt{2} - 1$. Multiplying by 2 we get $2(\sqrt{2} - 1)$.

As $4 < 8 < 9$, we have $2 < 2\sqrt{2} < 3$ and $0 < 2(\sqrt{2} - 1) < 1$. The integer part is 0 and the fractional part is $2(\sqrt{2} - 1)$. Multiplying by 2 we get $4(\sqrt{2} - 1)$.

As $25 < 32 < 36$, we have $5 < 4\sqrt{2} < 6$ and $1 < 4(\sqrt{2} - 1) < 2$. The integer part is 1 and the fractional part is $4\sqrt{2} - 5$. Multiplying by 2 we get $8\sqrt{2} - 10$.

As $121 < 128 < 144$, we have $11 < 8\sqrt{2} < 12$ and $1 < 8\sqrt{2} - 10 < 2$. The integer part is 1 and the fractional part is $8\sqrt{2} - 11$.

We have $(\sqrt{2})_{10} = (1.011...)_2$

4. The following questions are related to the base 2 and base 3 representation of numbers.

(a) A magician shows the audience the following cards:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Card 1 | 1 | 3 | 5 | 7 | 9 | 11 | 13 | 15 |
| Card 2 | 2 | 3 | 6 | 7 | 10 | 11 | 14 | 15 |
| Card 3 | 4 | 5 | 6 | 7 | 12 | 13 | 14 | 15 |
| Card 4 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |

The audience secretly picks a number between 1 and 15 inclusive, and then hands the magician the cards on which this number appears. The magician is then able to tell the audience right away what that number is. For example, if the audience's number is on cards 1, 3 and 4 but not card 2, the magician can compute the number by adding the first numbers on cards 1, 3 and 4, i.e. $1 + 4 + 8 = 13$. Explain how this trick works. (Hint: base 2 representation).

(b) Among 9 coins, there are 8 real ones and a fake one. All real coins have the same weight, which is greater than the weight of the fake coin. In two weighing on a balance, identify the fake coin. (Hint: base 3 representation).

**Solution**

(a) Consider the binary representation of a number between 1 and 15 inclusive, written as $(ABCD)_2$. If a number appears on card 1, $D = 1$. Else, $D = 0$. If a number appears on card 2, $C = 1$. Else, $C = 0$. If a number appears on card 3, $B = 1$. Else, $B = 0$. And if a number appears on card 4, $A = 1$. Else, $A = 0$. The number can be computed as $8.A + 4.B + 2.C + 1.D$. The smallest number on card 1 is 1, the smallest number on card 2 is 2, the smallest number on card 3 is 4 and the smallest number on card 4 is 8. Thus summing the smallest numbers on the cards where a number appears indeed gives the number. For example, if the audience's number is on cards 1, 3 and 4 but not on card 2, the number is $(1101)_2 = 8 + 4 + 1 = 13$.

(b) In a weighing, there are three possible outcomes: either pan can go down, or there may be equilibrium. For this reason, we use base 3, where the labels from 0 to 8 become 00, 01, 02, 10, 11, 12, 20, 21 and 22, with leading 0s inserted. We shall determine the label of the fake coin as follows. In the first weighing, place the three coins with 0 as the first ternary digit of their labels in one pan and those with 2 as the first ternary digit of their labels in other other. If the first pan goes down, 2 is the first ternary digit of the label of the fake coin. If the second pan goes down, 0 is the first ternary digit of the label of the fake coin. If we have equilibrium, then the first ternary digit of the label of the fake coin is 1. In the second weighing, we look at the second ternary digit of the labels, and again put the three 0s up against the three 2s. We can determine the second ternary digit of the label of the fake coin as in the first weighing. This will uniquely identify the fake coin in two weighing.